

# SSL CERTIFICATE CHECKER – PROJECT SUMMARY

## Overview

This project is a Python-based SSL Certificate Checker developed to analyze SSL/TLS certificates of websites and determine their validity, issuer information, and expiration status.

The tool establishes a secure connection with a target website, retrieves its SSL certificate, extracts certificate details, and calculates the remaining validity period. This helps identify expired or soon-to-expire certificates that could impact website security and trust.

## Features Added

- Fixed SSL certificate parsing issues
- Professional ASCII banner interface
- Certificate issuer information extraction
- SSL certificate validity verification
- Certificate expiration date detection
- Remaining validity days calculation
- Domain validation support
- Enhanced exception handling
- User-friendly terminal output
- Secure TLS connection establishment

## Certificate Information Checked

- Certificate Issuer
- Certificate Valid From Date
- Certificate Expiration Date
- Certificate Validity Status
- Remaining Days Until Expiry
- SSL/TLS Handshake Verification
- Domain Certificate Validation

## Technologies Used

- Python
- SSL Module
- Socket Programming
- Datetime Module

## Example Output

```
(root@kali)-[/home/kali/Desktop]
# python ssl_analyzer.py

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SSL Certificate Checker

Enter Domain (e.g. google.com): kochinc.com

=====
SSL CERTIFICATE INFORMATION
=====
Domain           : kochinc.com
Issuer            : Amazon
Valid From        : Mar 31 00:00:00 2026 GMT
Valid Until       : Oct 14 23:59:59 2026 GMT
Days Remaining    : 127
Status            : Valid
=====
```

## Concepts Demonstrated

- SSL/TLS Fundamentals
- Digital Certificates
- Public Key Infrastructure (PKI)
- Secure Socket Programming
- Certificate Validation
- Network Security Assessment
- Python Automation
- Error Handling

## Learning Outcome

This project helps understand how SSL/TLS certificates secure online communications and protect users against threats such as:

- Man-in-the-Middle (MITM) Attacks
- Data Interception
- Website Spoofing
- Certificate Expiration Risks
- Unencrypted Communications
- Identity Verification Issues

## Future Enhancements

- Display Subject Alternative Names (SAN)
- Detect TLS Version
- Show Cipher Suite Information
- Generate PDF Reports
- Export Results to JSON/CSV
- Bulk Domain Certificate Scanning
- WHOIS Integration
- Certificate Chain Validation

### **Disclaimer**

This project is intended for educational purposes and authorized security assessments only. Always ensure you have permission before performing security testing on any target system.

30 Days of Cybersecurity Automation – Day 06

Building practical cybersecurity tools through hands-on automation and real-world security assessments.

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